

# **Evaluation of Antidiabetic Activity of *Clitoria ternatea* Extract in a Glucose-Induced Zebrafish Embryo Model**

## **A MINI PROJECT REPORT**



Submitted by

**RUTH BODO D (922524214023)**

*in partial fulfillment for the award of the degree of*

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*In*

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**DEPARTMENT OF BIOTECHNOLOGY  
V.S.B. ENGINEERING COLLEGE, KARUR.  
(AN AUTONOMOUS INSTITUTION)  
ANNA UNIVERSITY: CHENNAI - 600025  
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**BONAFIDE CERTIFICATE**

Certified that this mini project report entitled " **Evaluation of Antidiabetic Activity of *Clitoria ternatea* Extract in a Glucose-Induced Zebrafish Embryo Model** " is the bonafide work of "**Ruth Bodo D (922524214023)**" who carried out the project work under my supervision during the academic year 2026 - 2027.

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**EXTERNAL EXAMINER**

## DECLARATION

I declare that the mini project report entitled "**Evaluation of Anti-Diabetic Activity of *Clitoria ternatea* Extract in Glucose Induced Zebra Fish Embryos**" submitted by me for the III Year Mini Project in Bachelor of Technology in Biotechnology is the original and independent work carried out under the guidance of Dr. D. Karthiga, Assistant Professor, Department of Biotechnology, V.S.B. Engineering College, Karur. This document has not previously formed the basis for the award of any degree, diploma, fellowship, or similar academic title.

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**RUTH BODO D**



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We endeavour to impart futuristic technical education of the highest quality to the student community and to inculcate discipline in them to face the world with self-confidence and thus we prepare them for life as responsible citizens to uphold human values and to be of service at large. We strive to bring of the Institution as an Institution of academic excellence of international standard.

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**PEO 1** - Successful professional career and/or higher studies by gaining knowledge in fundamental mathematics and biological principles.

**PEO 2** - Provide strong foundation in the core biotechnology courses to evaluate real life problems and to propose biotechnological solutions with economical and social viability

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The Biotechnology Graduates will have the ability to

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Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

### **PO3: Design/Development of Solutions**

Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

### **PO4: Conduct Investigations of Complex Problems**

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Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

**PO8: Ethics**

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**PO9: Individual and Team Work**

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**PO10: Communication**

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**PO11: Project Management and Finance**

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**PSO II:** Able to expose their skills using latest tools to arrive cost effective and appropriate solutions.

**PSO III:** Apply the contextual knowledge with professional ethics to manage different projects in a multi-disciplinary environment.

## ABSTRACT

Diabetes mellitus is a chronic metabolic disorder characterized by persistent hyperglycaemia and is a major global health concern due to its increasing prevalence and associated complications. Although several synthetic antidiabetic drugs are available, their long-term use may cause adverse effects, creating a need for safer and more effective plant-based therapeutic alternatives. *Clitoria ternatea* (butterfly pea) is a medicinal plant known for its rich phytochemical composition, including flavonoids, anthocyanins, and phenolic compounds, which have been reported to possess antioxidant and antidiabetic properties. The present study aims to evaluate the antidiabetic activity of *Clitoria ternatea* flower extract using a glucose-induced zebrafish embryo model. Zebrafish embryos serve as an effective experimental model due to their genetic similarity to humans, rapid development, optical transparency, and suitability for drug screening. This study is expected to provide scientific evidence regarding the antidiabetic potential of *Clitoria ternatea* and contribute to the development of safe and effective plant-based therapies for diabetes management.

**Keywords:** Diabetes mellitus; *Clitoria ternatea*; Zebrafish embryo (*Danio rerio*); Antidiabetic activity; Glucose-induced hyperglycaemia; Medicinal plants.

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## LIST OF SYMBOLS AND ABBREVIATIONS

| Abbreviation / Symbol | Expansion / Meaning                                    |
|-----------------------|--------------------------------------------------------|
| CT                    | <i>Clitoria ternatea</i>                               |
| DM                    | Diabetes Mellitus                                      |
| T2DM                  | Type 2 Diabetes Mellitus                               |
| GC-MS                 | Gas Chromatography–Mass Spectrometry                   |
| WHO                   | World Health Organization                              |
| OECD                  | Organisation for Economic Co-operation and Development |
| ROS                   | Reactive Oxygen Species                                |
| hpf                   | Hours Post Fertilization                               |
| dpf                   | Days Post Fertilization                                |
| DNA                   | Deoxyribonucleic Acid                                  |
| RNA                   | Ribonucleic Acid                                       |
| $\alpha$ -Amylase     | Alpha-Amylase                                          |
| $\alpha$ -Glucosidase | Alpha-Glucosidase                                      |
| °C                    | Degree Celsius                                         |
| %                     | Percentage                                             |
| g                     | Gram                                                   |
| mg                    | Milligram                                              |
| kg                    | Kilogram                                               |

|                  |                          |
|------------------|--------------------------|
| $\mu\text{g}$    | Microgram                |
| mL               | Millilitre               |
| $\mu\text{L}$    | Microlitre               |
| mg/mL            | Milligram per Millilitre |
| $\mu\text{g/mL}$ | Microgram per Millilitre |
| $\mu\text{M}$    | Micromolar               |
| h                | Hour                     |
| min              | Minute                   |



# 1. INTRODUCTION

## 1.1 Diabetes Mellitus

Diabetes mellitus (DM) is a chronic metabolic disorder characterized by persistent hyperglycemia resulting from impaired insulin secretion, insulin action, or both. Insulin, a peptide hormone produced by the pancreatic  $\beta$ -cells, plays a vital role in regulating blood glucose homeostasis by facilitating cellular glucose uptake and utilization. Impaired insulin function leads to elevated blood glucose levels, which, if left uncontrolled, can result in long-term damage to multiple organs, particularly the eyes, kidneys, nerves, heart, and blood vessels. According to the World Health Organization, 14% of adults aged 18 years and older were living with diabetes in 2022, an increase from 7% in 1990, and more than half (59%) of adults aged 30 years and over living with diabetes were not taking medication for their diabetes in that year. Diabetes treatment coverage was lowest in low- and middle-income countries. In 2021, diabetes was the direct cause of 1.6 million deaths, with 47% of all diabetes-related deaths occurring before age 70; an additional 530,000 deaths were caused by diabetes-related kidney disease, and elevated blood glucose accounted for roughly 11% of cardiovascular deaths. Among the different forms of diabetes, Type 2 diabetes mellitus (T2DM) accounts for more than 90–95% of all diagnosed cases. It is primarily characterized by insulin resistance accompanied by progressive pancreatic  $\beta$ -cell dysfunction, resulting in chronic hyperglycemia. Major risk factors for T2DM include obesity, physical inactivity, unhealthy dietary habits, ageing, and genetic predisposition. Persistent hyperglycemia promotes oxidative stress through excessive production of reactive oxygen species (ROS), leading to cellular injury and the development of chronic complications such as diabetic nephropathy, retinopathy, neuropathy, and cardiovascular disease. The prevalence of T2DM has increased rapidly over recent decades, making it one of the leading causes of morbidity and mortality worldwide. Although several synthetic antidiabetic drugs are currently available, prolonged use may be associated with adverse effects, high treatment costs, and reduced therapeutic efficacy in some patients. Consequently, increasing attention has been directed toward medicinal plants containing bioactive phytochemicals with antioxidant and antihyperglycemic properties as potential alternative or complementary therapies for diabetes management.

## 1.2 *Clitoria ternatea*

*Clitoria ternatea* L., commonly known as butterfly pea, is a perennial climbing legume belonging to the family Fabaceae. It is native to tropical Asia and is widely distributed across India, Southeast Asia, Africa, and Central and South America. In India, it is commonly known as Aparajita (Hindi and Bengali) and Kakkattan (Tamil). The plant has been extensively used in

traditional systems of medicine, including Ayurveda, for the treatment of various ailments such as inflammation, fever, skin diseases, memory disorders, and diabetes. *C. ternatea* is rich in several bioactive phytochemicals, including flavonoids, anthocyanins (ternatins), phenolic compounds, alkaloids, saponins, tannins, terpenoids, glycosides, steroids, and cyclotides. The blue petals in particular contain phenolic anthocyanins known as ternatins, along with the flavonoids kaempferol, quercetin, and myricetin, and these phytochemicals collectively exhibit antioxidant, anti-inflammatory, antimicrobial, neuroprotective, hepatoprotective, and antidiabetic effects. Several experimental and clinical studies have investigated the antidiabetic mechanisms of *C. ternatea*. Its antidiabetic potential is largely attributed to the antioxidant activity of its anthocyanins and flavonoids, which protect pancreatic  $\beta$ -cells from oxidative stress, along with their interaction with  $\alpha$ -amylase, a key digestive enzyme responsible for carbohydrate hydrolysis — inhibiting this enzyme delays intestinal glucose release and lowers postprandial blood glucose. Bioactive compounds such as flavonoids, anthocyanins, and alkaloids have also been reported to improve insulin production and enhance cellular insulin sensitivity. Consistent with this, a randomized crossover trial in healthy human subjects reported that *Clitoria ternatea* extract suppressed the peak postprandial glucose and insulin response after sucrose consumption, an effect attributed to inhibition of intestinal  $\alpha$ -glucosidase and pancreatic  $\alpha$ -amylase. Previous animal studies have similarly reported that administration of *C. ternatea* extracts reduced fasting blood glucose levels and improved antioxidant enzyme activity in diabetic rodent models. These findings suggest that the plant possesses considerable therapeutic potential for diabetes management. Therefore, *C. ternatea* has emerged as a promising natural source of antidiabetic compounds and warrants further investigation using reliable experimental models such as the glucose-induced zebrafish embryo model.

### 1.3 Zebrafish Embryo Model

The zebrafish (*Danio rerio*) has emerged as a valuable vertebrate model organism in biomedical research due to its genetic similarity to humans, rapid embryonic development, optical transparency, high fecundity, and ease of maintenance. Approximately 70% of human genes have at least one zebrafish ortholog, and 84% of genes known to be linked with human diseases have zebrafish counterparts, and many metabolic pathways involved in glucose homeostasis and pancreatic development are highly conserved between zebrafish and humans. These characteristics make zebrafish an excellent model for investigating the pathophysiology of metabolic disorders, including diabetes mellitus. Zebrafish embryos are particularly advantageous for biomedical research because they develop externally, allowing direct

observation of embryogenesis and organ development without invasive procedures. Their transparent embryos and larvae facilitate real-time visualization of metabolic and vascular changes, including pancreatic development and glucose-induced abnormalities, using light or fluorescence microscopy. Furthermore, their small size, rapid development, and high reproductive capacity enable high-throughput screening of therapeutic compounds targeting obesity-related metabolic syndrome in a cost-effective and time-efficient manner. In diabetes research, glucose-induced zebrafish embryo models have been widely employed to investigate hyperglycemia, oxidative stress, insulin resistance, pancreatic  $\beta$ -cell dysfunction, and glucose metabolism. Exposure of zebrafish embryos to elevated glucose concentrations induces hyperglycemic conditions that resemble several pathological features of type 2 diabetes mellitus in humans. For example, exposure of zebrafish embryos to high glucose has been shown to cause growth retardation and craniofacial cartilage malformations linked to reactive oxygen species accumulation and oxidative stress, while separate work has shown that glucose-induced developmental toxicity in zebrafish larvae involves ROS accumulation, disrupted antioxidant system function, and mitochondria-mediated apoptosis. This experimental model provides an efficient platform for evaluating the antidiabetic efficacy of natural products, synthetic compounds, and novel drug candidates. In addition to their biological relevance, zebrafish embryos offer ethical advantages over conventional mammalian models during their early stages of development, since embryos and early larvae generally fall outside standard animal-welfare licensing requirements in many jurisdictions. Their conserved physiological responses and suitability for in vivo drug screening have established zebrafish as a reliable and widely accepted model for studying diabetes and assessing the therapeutic potential of medicinal plant extracts.

Therefore, the glucose-induced zebrafish embryo model is considered an appropriate and reliable experimental system for evaluating the antidiabetic activity of *Clitoria ternatea* extract and for understanding its potential protective effects against glucose induced metabolic alterations.

#### **1.4 Rationale for Selecting the Zebrafish Model**

While rodent models have traditionally been used to evaluate the antidiabetic potential of medicinal plant extracts, they present notable limitations for early-stage screening, including high maintenance costs, long experimental timelines, low offspring numbers, and stricter ethical and regulatory requirements; these constraints limit the number of extracts, fractions, or doses that can be feasibly tested within a reasonable timeframe, a particular disadvantage when screening a botanical extract like *Clitoria ternatea*, whose activity may vary with solvent, plant part, and concentration. Zebrafish address several of these limitations: they demonstrate

remarkable translational value for high-throughput screening of therapeutic compounds targeting metabolic disorders and for validating conserved molecular pathways underlying insulin resistance and  $\beta$ -cell dysfunction, while their pancreatic architecture — featuring conserved islet organization and  $\beta$ -cell regulatory mechanisms — helps address limitations inherent to rodent models. Given these advantages, together with the conserved glucoseregulatory pathways described above, the zebrafish embryo offers a practical and physiologically relevant intermediate model for screening *C. ternatea* extract before, or alongside, confirmatory studies in mammalian systems

### **1.5 Research Hypothesis**

*Clitoria ternatea* flower extract possesses significant antidiabetic activity and can reduce glucose-induced metabolic alterations in zebrafish embryos.

### **1.6 Aim of the present study**

The present study was undertaken to evaluate the antidiabetic potential of *Clitoria ternatea* flower extract using a glucose-induced zebrafish (*Danio rerio*) embryo model. The study aims to investigate the protective effects of the extract against glucose induced metabolic alterations and to assess its potential as a natural antidiabetic agent. The findings are expected to provide experimental evidence supporting the therapeutic potential of *Clitoria ternatea* and serve as a foundation for future preclinical and translational studies.

## 2. REVIEW OF LITERATURE

### 2.1 Previous Studies on Medicinal Plants in Diabetes Management

Medicinal plants have attracted considerable attention as potential therapeutic agents for diabetes mellitus because of their diverse bioactive constituents and relatively fewer adverse effects compared with synthetic drugs. Numerous medicinal plants contain flavonoids, alkaloids, phenolic compounds, tannins, saponins, and terpenoids that exhibit anti-hyperglycaemic, antioxidant, and anti-inflammatory properties. These phytochemicals regulate blood glucose levels through multiple mechanisms, including stimulation of insulin secretion, enhancement of insulin sensitivity, inhibition of carbohydrate-digesting enzymes, and protection of pancreatic  $\beta$ -cells from oxidative stress. A systematic review by Alam et al. (2022) reported that plant-derived compounds significantly improved glycaemic control in experimental diabetic models and highlighted their potential as complementary therapies for diabetes management. Similarly, Salehi et al. (2019) emphasized that naturally occurring polyphenols possess strong antioxidant activity capable of reducing oxidative stress, which plays a major role in the development of diabetic complications. These findings have encouraged researchers to investigate traditional medicinal plants as promising alternatives for the development of safer antidiabetic drugs.

### 2.2 Previous Studies on the Antidiabetic Activity of *Clitoria ternatea*

Among the medicinal plants investigated for diabetes management, *Clitoria ternatea* has gained considerable attention because of its rich phytochemical composition and long history of use in traditional medicine. Several studies have demonstrated that extracts prepared from different parts of the plant exhibit significant anti-hyperglycaemic activity. Chusak et al. (2018) conducted a randomized crossover study in healthy volunteers and observed that consumption of butterfly pea flower extract together with sucrose significantly reduced postprandial blood glucose and insulin responses. The authors suggested that inhibition of  $\alpha$ -amylase and  $\alpha$ -glucosidase enzymes, together with the antioxidant activity of anthocyanins, contributed to these beneficial effects.

Experimental animal studies have also reported encouraging results. Administration of *Clitoria ternatea* flower extract to streptozotocin-induced diabetic rats resulted in significant reductions in fasting blood glucose levels together with improvements in antioxidant enzyme activity and pancreatic tissue architecture. Researchers have suggested that flavonoids and anthocyanins present in the extract reduce oxidative stress, preserve pancreatic  $\beta$ -cell integrity, and improve insulin secretion. In addition, several in vitro investigations have demonstrated that *Clitoria*

*ternatea* possesses strong free radical scavenging activity, further supporting its potential role in preventing diabetes-associated oxidative damage. Although these findings demonstrate the therapeutic potential of *Clitoria ternatea*, most published studies have been limited to cell-based assays and mammalian diabetic models.

## **2.3 Previous Studies Using Zebrafish Embryos in Diabetes Research**

The zebrafish (*Danio rerio*) has become an important vertebrate model for studying metabolic disorders because many genes and signalling pathways involved in glucose metabolism are highly conserved between zebrafish and humans. Its transparent embryos, rapid development, and ease of maintenance make it particularly suitable for evaluating drug toxicity and therapeutic efficacy. Researchers have successfully developed glucose-induced zebrafish embryo models that mimic several pathological features of type 2 diabetes mellitus, including hyperglycaemia, oxidative stress, impaired pancreatic development, and metabolic dysfunction. Studies have demonstrated that exposure of zebrafish embryos to elevated glucose concentrations results in developmental abnormalities, increased production of reactive oxygen species, and alterations in glucose homeostasis. Jurczyk et al. reported that zebrafish exhibit conserved mechanisms regulating pancreatic  $\beta$ -cell development and insulin signalling, making them valuable for diabetes research. Likewise, other investigators have shown that glucose-induced zebrafish embryos provide a rapid and cost-effective platform for screening natural products with potential antidiabetic activity. Compared with conventional rodent models, zebrafish require less space, lower maintenance costs, and smaller quantities of test compounds, enabling efficient high-throughput screening of medicinal plant extracts.

## **2.4 Research gap**

Although several studies have reported the antidiabetic activity of *Clitoria ternatea* using in vitro enzyme inhibition assays and diabetic rodent models, limited information is available regarding its efficacy in glucose-induced zebrafish embryo models. Most published research has focused primarily on blood glucose reduction, antioxidant activity, and histopathological changes in mammals, while comparatively little attention has been given to evaluating the protective effects of *Clitoria ternatea* during early vertebrate development. Considering the increasing use of zebrafish embryos as an alternative model for drug screening and toxicity assessment, further investigation is required to establish the antidiabetic potential of *Clitoria ternatea* in this model. Therefore, the present study was undertaken to evaluate the protective effects of *Clitoria ternatea* flower extract against glucose-induced metabolic alterations using zebrafish embryos.

### 3. MATERIALS AND METHODS

#### 3.1 Collection and authentication of plant material

The flowers of *Clitoria ternatea* were collected from Gandhigramam, Karur district, Tamil Nadu  
10.95



#### 3.1 Shade dried *Clitoria ternatea*



Authenticated by

*Amj*  
The Director  
THE RAPINAT HERBARIUM and  
CENTRE FOR MOLECULAR SYSTEMATICS  
St. Joseph's College (Autonomous)  
Tiruchirappalli - 620 002,  
TAMILNADU, INDIA.

M.S.001

கல்வி நிறுவனத்தின் பெயர் : Trichy Research Institute of Biotechnology  
Name of the Institution

பெயர் : *Clitoria ternatea L.*  
Binominal Name

குடும்பம் : *Fabaceae*  
Family

பெயர் : *Butterfly Pea*  
Common Name

சேகரித்தவரின் பெயர் : *Ruth Bode D, Monica Sree V, Sandhya*  
Name of the person collected

சேகரித்த இடம் : *Gandhigramam, Karur*  
Place of the collection

சேகரித்த நாள் : *07/07/2026*  
Date of collection

காம்பெனேட்டர்ஸ் மனவை 99 6 55 62 385

### 3.2 Authentication of *Clitoria ternatea*



### 3.2 Soxhlet extraction of *Clitoria ternatea* flower using hydro alcohol

Soxhlet extraction is a very useful tool for preparative purposes in which the analyte is concentrated from the matrix as a whole or separated from particular interfering substances. Sample preparation of environmental samples has been developed for decades using a wide variety of techniques. Solvent extraction of solid samples, which is commonly known as solid–liquid extraction (also referred to as leaching or Lixiviation in a more correct use of the physicochemical terminology), is one of the oldest methods for solid sample pre-treatment. Conventional Soxhlet extraction remains as one of the most relevant techniques in the environmental extraction field.

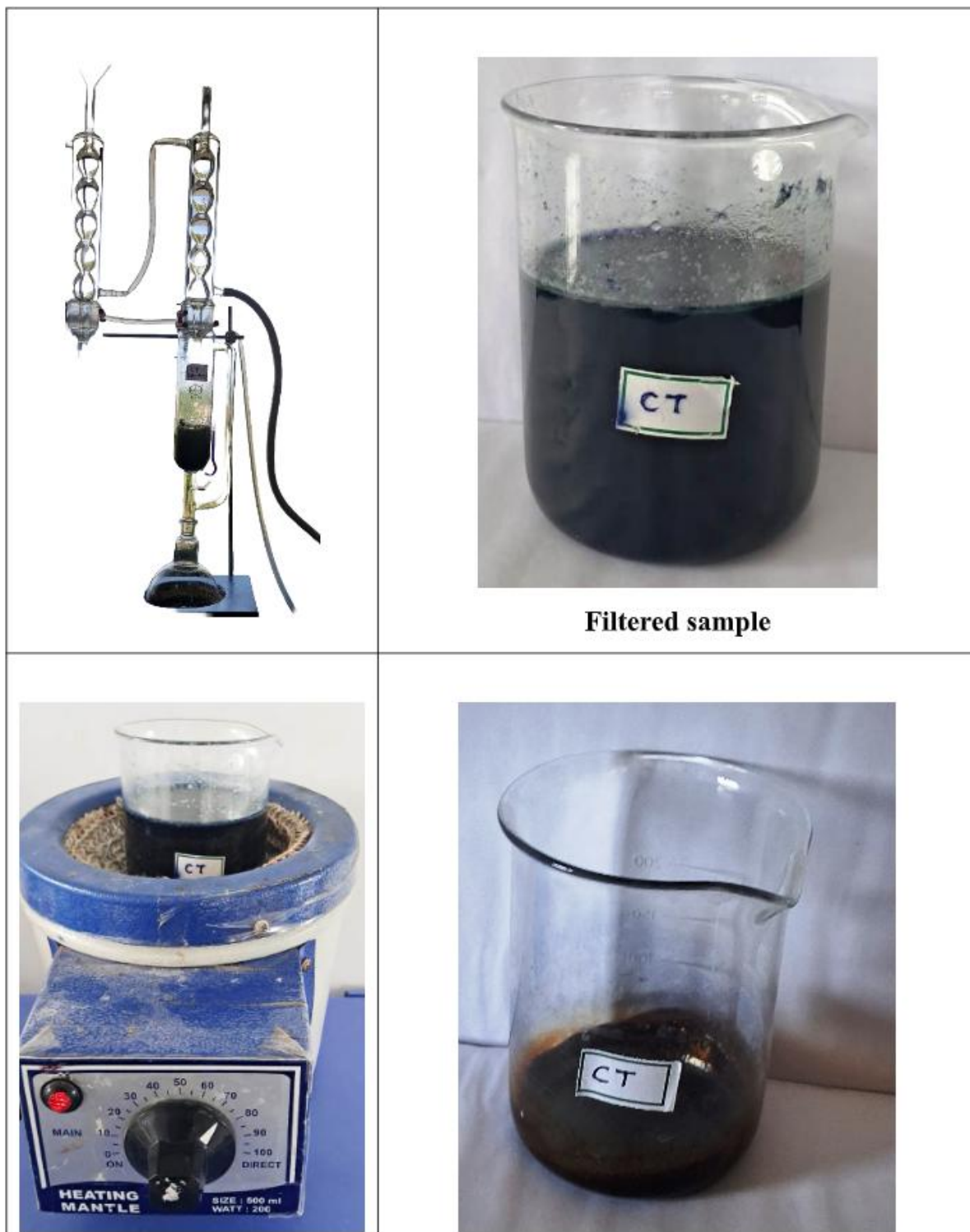
#### Materials Required

Ethanol was purchased from Merk, USA. Whatman No.1 filter paper was purchased from Millipore, USA.

#### Procedure

1. Test sample (CT) can be fresh or dried. It needs to be crushed, using a pestle and mortar, to provide a greater surface area.
2. The test sample should be sufficient to fill the porous cellulose thimble (in our experiments we use an average of 14 g of thyme in a 25- x 80-mm thimble).
3. All equipment should be too assembled. Build a rig using stands and clamps to support the extraction apparatus.
4. Following this, 20g of sample CT and 250ml of Hydro alcohol added to a round bottom flask, which is attached to a Soxhlet extractor and condenser on heating mantle.
5. The crushed plant material is loaded into the thimble, which is placed inside the Soxhlet extractor.
6. The side arm is lagged with glass wool.
7. The solvent is heated using the heating mantle and will begin to evaporate, moving through the apparatus to the condenser.

8. The condensate then drips into the reservoir containing the thimble.
9. Once the level of solvent reaches the siphon it pours back into the flask and the cycle begins again.
10. The process should run for a total of 4 hours.
11. Once the extraction set up, it can be left to run without direct supervision.
12. It is not advised to leave the equipment completely alone due to the mix of running water and an electrical appliance, so a technician or other lab user should be made aware.
13. The equipment can be turned off.



### 3.3 preparation of Soxhlet extraction of *Clitoria ternatea*

| Heated at 70° C                                                                                                          | After Evaporation |
|--------------------------------------------------------------------------------------------------------------------------|-------------------|
|  <p><b>Final product of sample</b></p> |                   |

#### 3.4 Storage of Soxhlet extraction of *Cutoria ternatea*

$$\text{Percentage of Yield obtained} = \frac{\text{Yield in grams}}{\text{Loaded sample in grams}} \times 100$$

$$\text{Percentage of Yield obtained} = ( 2.45 / 20 ) \times 100$$

$$\text{Percentage of Yield obtained} = 12.25 \%$$

### 3.3 Embryo toxicity study

The toxicity activity was assessed by the procedure of WHO with some modification. For the busy test, Zebra fish embryo were taken in five batches along with the control. The numbers of dead Zebra fish embryo were counted after treated exposure of CT sample and the percentage of mortality was reported from the average of five replicates. The percentage of mortality was analysed using the following formula was calculated using Graph Pad Prism software (USA).

## Preparation of Stock and working solution

Stock Solution:

Take 50mg of plant extract and add 1ml of solvent (water or ethanol).

Working Solution:

Take 100  $\mu\text{g}$  from the stock solution and add 900  $\mu\text{g/mL}$  distilled water.

Different concentration:

- i) 500  $\mu\text{g/mL}$
- ii) 250  $\mu\text{g/mL}$
- iii) 100  $\mu\text{g/mL}$
- iv) 50  $\mu\text{g/mL}$
- v) 10  $\mu\text{g/mL}$
- vi) Control

## Procedure

1. Adult healthy zebrafish (*Danio rerio*) were maintained under standard laboratory conditions to obtain fertilized eggs through natural spawning.
2. Fertilized embryos were collected within 2–4 hours post-fertilization (hpf) and examined under a stereomicroscope. Unfertilized, damaged, or abnormal embryos were discarded, and only healthy embryos were selected for the experiment.
3. Stock solutions of *Clitoria ternatea* flower extract were prepared, and different test concentrations were obtained by dilution with embryo medium.
4. Healthy embryos were randomly distributed into 24-well or 96-well culture plates, with one embryo placed in each well containing the appropriate volume of test solution. A control group containing embryo medium without plant extract was included.
5. The embryos were exposed to the different concentrations of the extract and incubated under standard zebrafish culture conditions (approximately 26–28.5°C) for up to 96 hours.
6. Embryos were observed every 24 hours under a stereomicroscope for mortality and developmental abnormalities.
7. Dead embryos were removed immediately to prevent contamination of the remaining embryos.
8. At the end of the exposure period, survival percentage, hatching rate, and morphological abnormalities were recorded and compared with the control group. Based on the toxicity observations, the highest non-toxic concentration of *Clitoria ternatea* extract was selected for

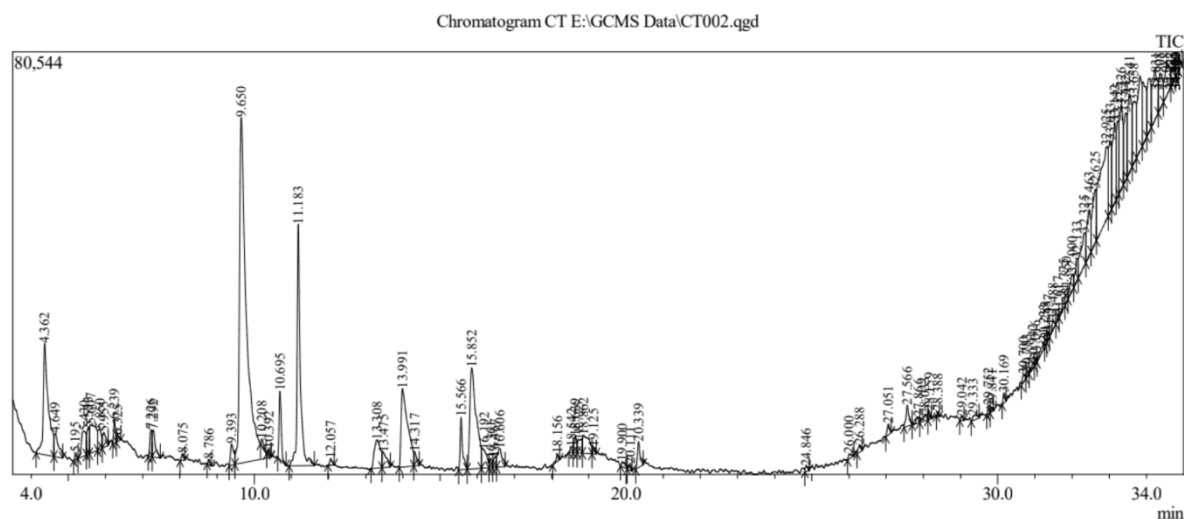
further evaluation of antidiabetic activity in the glucose-induced zebrafish embryo model.

### **3.4 Effect of *Clitoria ternatea* extract on diabetes induced zebra fish larvae**

1. Collect healthy zebrafish larvae at 5 days post-fertilization (5 dpf) and maintain them fresh water
2. Induce diabetes by exposing the larvae to the required concentration of glucose solution for 24 hours.
3. Divide the larvae into normal control, disease control (Glucose induced), standard drug-treated (Metformin), and *Clitoria ternatea* extract-treated groups.
4. Treat the larvae with different concentrations of *Clitoria ternatea* extract and incubate at  $28 \pm 0.5^{\circ}\text{C}$ .
5. Replace the treatment solution with fresh water
6. Observe the larvae under a microscope.
7. Capture images of representative larvae for comparison between groups.
8. Analyze the observations and compare the effects of *Clitoria ternatea* extract with the diabetic and control groups.

## 4.RESULTS AND DISCUSSION FRAMEWORK

### 4.1 GCMS chromatogram of *Clitoria ternatea*



### 4.2 Phytochemicals present in *Clitoria ternatea*

Peak Report TIC

| Peak# | R.Time | Area   | Height | Name                                            |
|-------|--------|--------|--------|-------------------------------------------------|
| 1     | 4.362  | 204965 | 21119  | 2-Cyclopenten-1-one, 2-hydroxy-                 |
| 2     | 4.649  | 34457  | 4439   | 2-Propyn-1-amine, N-methyl-                     |
| 3     | 5.195  | 3835   | 1277   | Pentanoic acid, 2-hydroxy-, ethyl ester         |
| 4     | 5.430  | 39804  | 4709   | Diethyl sulfone                                 |
| 5     | 5.533  | 24086  | 5029   | Allantoic acid                                  |
| 6     | 5.607  | 63043  | 5451   | Propanal, 3-methoxy-                            |
| 7     | 5.850  | 24983  | 3747   | Carbonic acid, heptyl phenyl ester              |
| 8     | 5.955  | 18872  | 2859   | 3-Butyn-2-amine, 2-methyl-                      |
| 9     | 6.239  | 14660  | 4391   | 1,5-Heptadiene-3,4-diol                         |
| 10    | 6.325  | 4884   | 1184   | n-Butyl nitrite                                 |
| 11    | 7.226  | 21079  | 5115   | Mequinol                                        |
| 12    | 7.292  | 26245  | 5084   | Butane, 2,2-dimethyl-                           |
| 13    | 8.075  | 5478   | 1274   | Cyclopropylamine                                |
| 14    | 8.786  | 2949   | 1018   | Propanamide, N-methyl-2-(methylamino)-          |
| 15    | 9.393  | 24604  | 3990   | 2-Hydroxy-gamma-butyrolactone                   |
| 16    | 9.650  | 864751 | 65839  | 4-Vinylphenol                                   |
| 17    | 10.208 | 6043   | 1406   | Methacrylamide                                  |
| 18    | 10.392 | 3842   | 1127   | Butane-1,3-dione, 4,4,4-trifluoro-2-phenyl-, 3- |
| 19    | 10.695 | 48816  | 12967  | 2H-Pyran-2-one, tetrahydro-6-propyl-            |
| 20    | 11.183 | 276661 | 46022  | 2-Methoxy-4-vinylphenol                         |
| 21    | 12.057 | 5704   | 1246   | n-Butyl nitrite                                 |
| 22    | 13.308 | 60395  | 5425   | Oxirane, [(2-propenyloxy)methyl]-               |
| 23    | 13.475 | 26708  | 2876   | 1-Butanol, 3,3-dimethyl-                        |
| 24    | 13.991 | 185897 | 14969  | .beta.-D-Glucopyranose, 1,6-anhydro-            |
| 25    | 14.317 | 15081  | 2708   | 2-Butanone, 1,1,1-trifluoro-                    |
| 26    | 15.566 | 44961  | 9963   | Diethyl Phthalate                               |
| 27    | 15.852 | 230675 | 19213  | 5-Methylhexanoic acid                           |
| 28    | 16.192 | 33955  | 3400   | Crotonic anhydride                              |
| 29    | 16.361 | 7342   | 1721   | cis-2-Nitro-4-t-butylcyclohexanone              |
| 30    | 16.467 | 11164  | 1949   | tert-Butyl cyclopropylmethyl sulfoxide          |
| 31    | 16.606 | 31322  | 3513   | Pentane, 2-methoxy-2,4,4-trimethyl-             |
| 32    | 18.156 | 6196   | 1438   | Difraactaic acid, 2TMS derivative               |
| 33    | 18.542 | 5816   | 1115   | 2-Butenoyl chloride                             |
| 34    | 18.639 | 14247  | 3110   | Difraactaic acid, 2TMS derivative               |

| Peak# | R.Time | Area    | Height | Name                                            |
|-------|--------|---------|--------|-------------------------------------------------|
| 35    | 18.725 | 21259   | 2530   | 1,3-Dioxan-4-one, 2-(1-methylethyl)-5-methyl    |
| 36    | 18.862 | 42306   | 3437   | 1,3-Dioxan-4-one, 2-(1-methylethyl)-5-methyl    |
| 37    | 19.125 | 6996    | 1346   | Propanoic acid, 2-(aminooxy)-                   |
| 38    | 19.900 | 8393    | 974    | 3-Butyn-2-ol                                    |
| 39    | 20.111 | 4978    | 1119   | n-Propyl acrylate                               |
| 40    | 20.339 | 26407   | 4885   | 1-(p-Toluidino)-1-deoxy-.beta.-d-idopyranose    |
| 41    | 24.846 | 5017    | 925    | Propargyl alcohol                               |
| 42    | 26.000 | 5045    | 937    | .beta.-Hydroxyethyltheophylline tert-butyldim   |
| 43    | 26.288 | 7257    | 1374   | 2-Methyl-5-hexen-3-ol                           |
| 44    | 27.051 | 9225    | 2096   | 2-Methylallyl methacrylate                      |
| 45    | 27.566 | 23471   | 4053   | Caprolactone oxime, (NB)-O-[(diethylborylox     |
| 46    | 27.866 | 7254    | 1166   | 4H-Pyran-4-one                                  |
| 47    | 28.033 | 10071   | 1065   | 2,4-Diamino-5-butyl-8-hydroxy-5H-chromeno       |
| 48    | 28.159 | 4914    | 1608   | Caprolactone oxime, (NB)-O-[(diethylborylox     |
| 49    | 28.388 | 2157    | 925    | Glycine, 2-cyclohexyl-N-(but-3-yn-1-yl)oxycar   |
| 50    | 29.042 | 3536    | 893    | Caprolactone oxime, (NB)-O-[(diethylborylox     |
| 51    | 29.333 | 8529    | 860    | 1,4-Dihydro-4-oxopyridazine                     |
| 52    | 29.752 | 5009    | 1562   | 1,4-Dihydro-4-oxopyridazine                     |
| 53    | 29.841 | 3354    | 923    | 2-Amino-1,3,5-triazine                          |
| 54    | 30.169 | 6610    | 1884   | Caprolactone oxime, (NB)-O-[(diethylborylox     |
| 55    | 30.700 | 7747    | 1272   | Thebaol, O-ethyl-                               |
| 56    | 30.783 | 3200    | 867    | 1-Propyl-3-piperidinamine, N-trifluoroacetyl-   |
| 57    | 30.942 | 6165    | 1105   | 1-Propanone, 1,1'-(2,4,6-trihydroxy-m-phenyle   |
| 58    | 31.016 | 2926    | 1114   | Benzene, 1-(1,3-dimethyl-2-butenyl)-4-(trifluo  |
| 59    | 31.233 | 11758   | 1329   | 4-Iodo-1H-pyrazol-5-amine                       |
| 60    | 31.283 | 3423    | 1157   | 4-Iodo-1H-pyrazol-5-amine                       |
| 61    | 31.357 | 4687    | 1450   | 1-Propanone, 1,1'-(2,4,6-trihydroxy-m-phenyle   |
| 62    | 31.488 | 8086    | 1707   | 2-Hexyne, 6-iodo-                               |
| 63    | 31.617 | 4009    | 1098   | 1-(3-Cyclohexen-1-ylmethyl)-4-piperidinamine    |
| 64    | 31.775 | 11393   | 2112   | Uridine, 2'-deoxy-, 3',5'-bis(trifluoroacetate) |
| 65    | 31.850 | 7273    | 1517   | 2H-Pyran, 5,6-dihydro-5-(p-methoxybenzoyl)-     |
| 66    | 32.000 | 20547   | 3177   | 1-(10,10-Dimethyl-3,3-dioxo-3-thia-4-azatricy   |
| 67    | 32.133 | 23833   | 4181   | 1,4-Bis(trimethylsilyl)benzene                  |
| 68    | 32.325 | 59108   | 6254   | (7-Methyl-2-thioxo-2H-[1,2,4]triazolo[1,5-a]p   |
| 69    | 32.463 | 62518   | 8666   | 3-Isopropylphenyl N-pentafluoropropionyl-N-     |
| 70    | 32.625 | 79339   | 10203  | 2,6-Dihydroxyacetophenone, 2TMS derivative      |
| 71    | 32.925 | 228949  | 13934  | 4-Isopropyl-2,6,7-trioxa-1-phosphabicyclo[2.2   |
| 72    | 33.033 | 72364   | 13519  | 1-Triethylsilyloxyheptadecane                   |
| 73    | 33.142 | 96769   | 15371  | (5-Ethylcyclopent-1-enyl)methanol               |
| 74    | 33.225 | 63363   | 14490  | 2,3-Dihydroxybenzoic acid, 3TMS derivative      |
| 75    | 33.326 | 108340  | 16005  | Hexasiloxane, 1,1,3,3,5,5,7,7,9,9,11,11-dodec   |
| 76    | 33.442 | 67535   | 12843  | 1-Triethylsilyloxyheptadecane                   |
| 77    | 33.541 | 114989  | 15000  | Pentasiloxane, dodecamethyl-                    |
| 78    | 33.658 | 79691   | 11807  | CJ 20556, 2TMS                                  |
| 79    | 33.831 | 111326  | 14235  | Nonadecanoic acid, benzyltrimethylsilyl ester   |
| 80    | 33.908 | 79493   | 11867  | 18-O-Feruloyloxyoctadec-9-enoic acid, methyl    |
| 81    | 34.088 | 71701   | 9973   | Succinic acid, dodec-2-en-1-yl 2,2,3,4,4,4-hex  |
| 82    | 34.227 | 85469   | 9479   | 2,6-Dihydroxybenzoic acid, 3TMS derivative      |
| 83    | 34.370 | 48713   | 6867   | 1,3,5-Triazine-2,4,6-(1H,3H,5H)-trione, 1,3,5-t |
| 84    | 34.551 | 54153   | 6282   | Heptasiloxane, 1,1,3,3,5,5,7,7,9,9,11,11,13,13  |
| 85    | 34.742 | 26318   | 4285   | 3,4-Dihydroxymandelic acid, ethyl ester, tri-T  |
| 86    | 34.820 | 17245   | 4292   | 18-O-Feruloyloxyoctadec-9-enoic acid, methyl    |
| 87    | 34.900 | 6134    | 1811   | 18-O-Feruloyloxyoctadec-9-enoic acid, methyl    |
|       |        | 4253872 | 534524 |                                                 |



### 4.3 Compounds with anti-diabetic activities

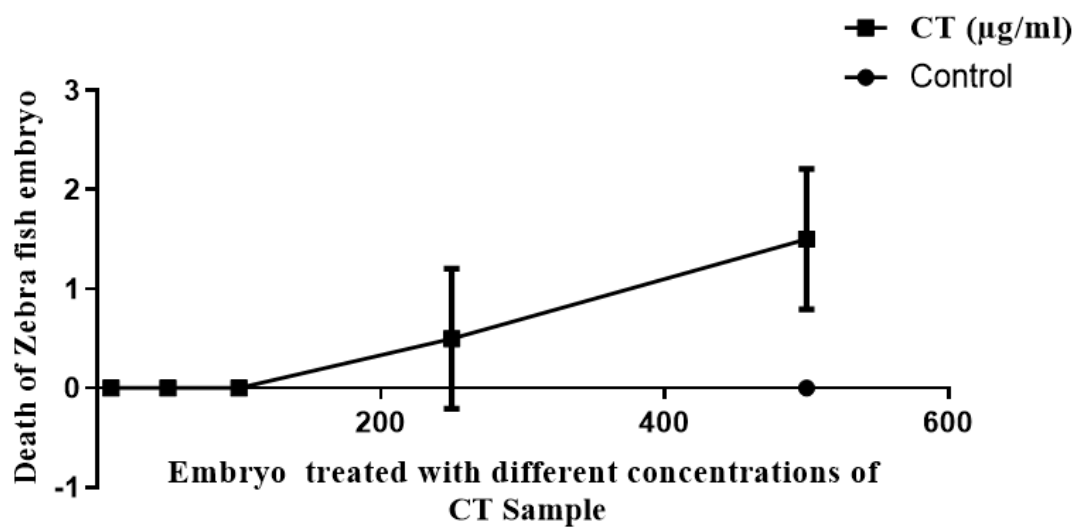
| Compound                                                    | Therapeutic Use                                              | Medicinal Use                                                                  |
|-------------------------------------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------|
| $\beta$ -D-Glucopyranose, 1,6-anhydro                       | Regulates glucose metabolism                                 | Supports blood glucose homeostasis and reduces glycation stress.               |
| 2,6-Dihydroxybenzoic acid (3TMS derivative)                 | Antioxidant; reduces oxidative stress                        | Protects pancreatic $\beta$ -cells and improves insulin sensitivity.           |
| 4-Vinylphenol                                               | Anti-inflammatory and antioxidant                            | Helps lower blood glucose levels and may prevent diabetic neuropathy.          |
| 2-Methoxy-4-vinylphenol                                     | Enhances insulin sensitivity; inhibits $\alpha$ -glucosidase | Reduces postprandial hyperglycemia.                                            |
| 5-Methylhexanoic acid                                       | Modulates lipid metabolism                                   | May reduce insulin resistance associated with fatty acid imbalance.            |
| 4H-Pyran-4-one                                              | Antioxidant; improves glucose metabolism                     | Reduces oxidative stress associated with diabetes.                             |
| Uridine derivatives (2'-deoxy forms)                        | Nucleoside analogues involved in metabolic regulation        | May provide $\beta$ -cell protection and support glucose regulation.           |
| Succinic acid esters                                        | Influences mitochondrial energy metabolism                   | Improves cellular energy balance under diabetic conditions.                    |
| 3,4-Dihydroxymandelic acid derivatives                      | Antioxidant; involved in catechol metabolism                 | Protects against oxidative damage associated with diabetes.                    |
| Feruloyl derivatives (18-O-Feruloyloxyoctadec-9-enoic acid) | Potent antioxidant and anti-inflammatory agent               | May reduce diabetic complications by controlling oxidative stress.             |
| Diethyl phthalate (plant-derived analogue)                  | Reported enzyme inhibitory activity                          | Investigated for $\alpha$ -glucosidase inhibitory activity in diabetes models. |
| Phenolic esters (Carbonic acid, heptyl phenyl ester)        | Antioxidant and anti-inflammatory                            | May contribute to improved glucose regulation.                                 |
| Crotonic anhydride derivatives                              | Anti-inflammatory                                            | Investigated for potential metabolic regulatory effects in diabetes.           |

D - Dead zebra fish embryos  
L - Live zebra fish embryos  
CT - *Clitoria ternatea*

L - Live zebra fish embryos

CT - *Clitoria ternatea*

| Name of the sample | Control |    | 500 µg/ml |   | 250 µg/ml |    | 100 µg/ml |    | 50 µg/ml |    | 10 µg/ml |    |
|--------------------|---------|----|-----------|---|-----------|----|-----------|----|----------|----|----------|----|
|                    | D       | L  | D         | L | D         | L  | D         | L  | D        | L  | D        | L  |
| CT                 | 0       | 10 | 2         | 8 | 1         | 9  | 0         | 10 | 0        | 10 | 0        | 10 |
|                    | 0       | 10 | 1         | 9 | 0         | 10 | 0         | 10 | 0        | 10 | 0        | 10 |

CT - *Clitoria ternatea*CT - *Clitoria ternatea*
$$\text{Percentage of mortality} = \frac{\text{Total N.O of Zebra fish egg}}{\text{Total N.O of Zebra fish egg}} \times 100$$

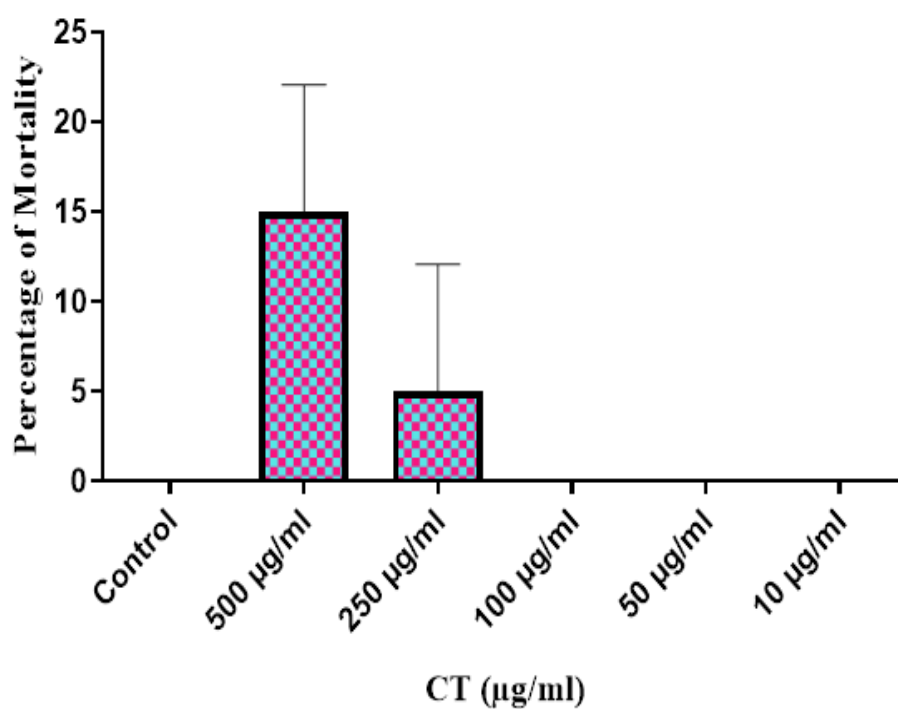
## 4.6 Calculation of percentage mortality

CT - *Clitoria ternatea*

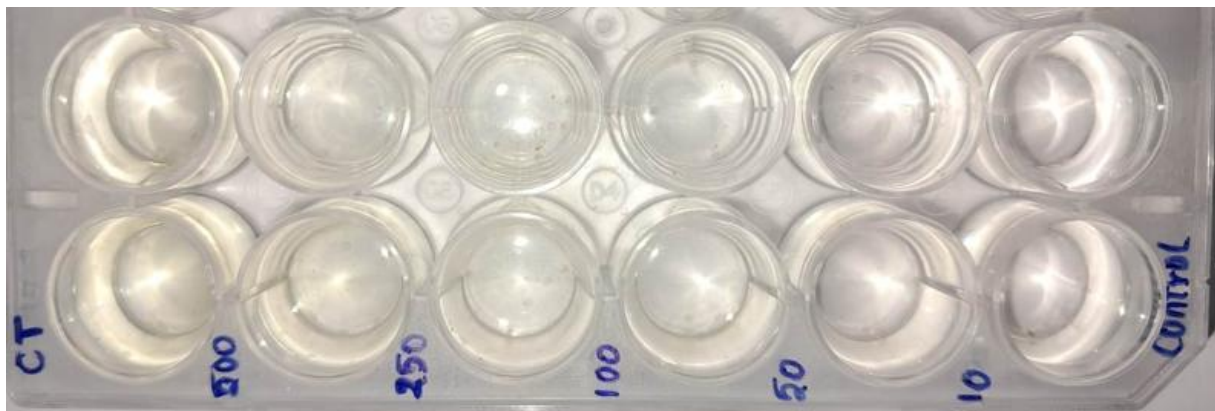
| Name of the sample | Control | 500 µg/ml | 250 µg/ml | 100 µg/ml | 50 µg/ml | 10 µg/ml |
|--------------------|---------|-----------|-----------|-----------|----------|----------|
| CT                 | 0       | 20        | 10        | 0         | 0        | 0        |
|                    | 0       | 10        | 0         | 0         | 0        | 0        |
| Mean value (%)     | 0       | 15        | 5         | 0         | 0        | 0        |

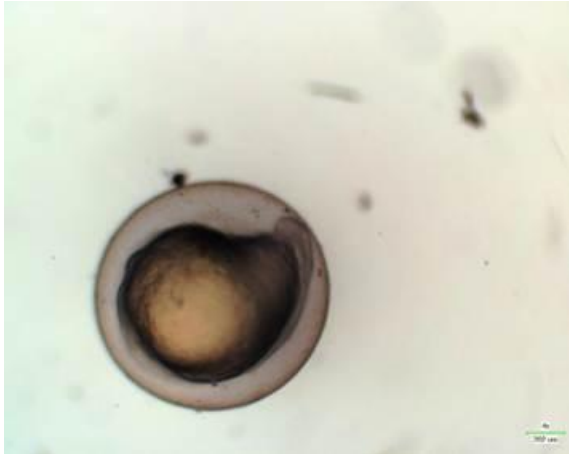
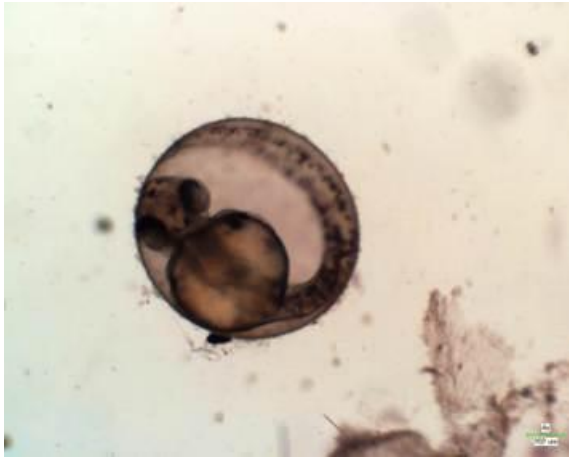
## 4.7 Percentage mortality Vs Concentration

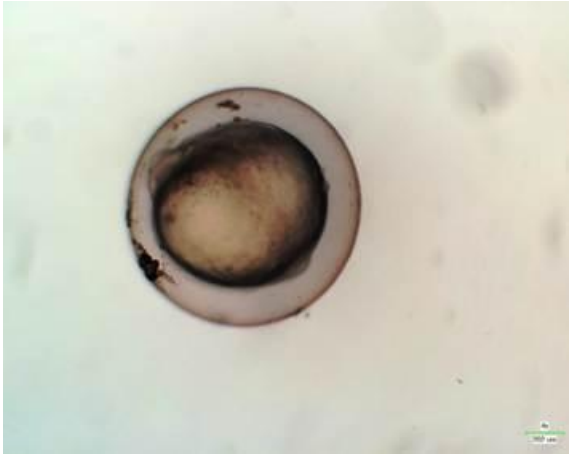
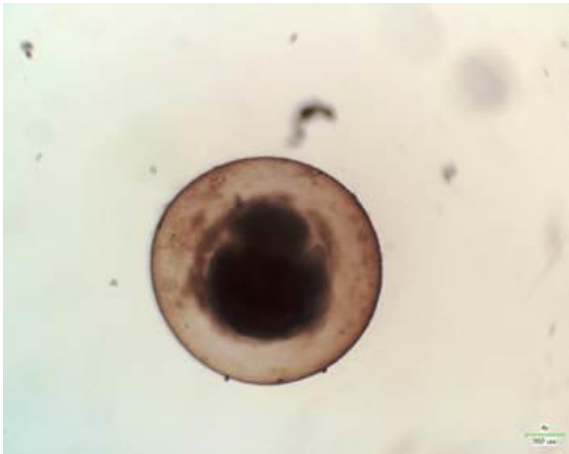
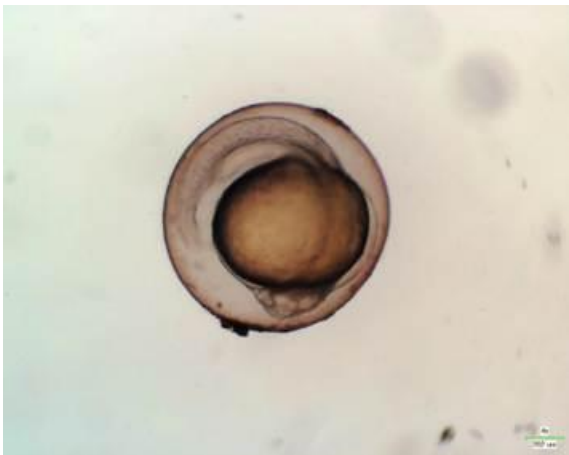
CT - *Clitoria ternatea*

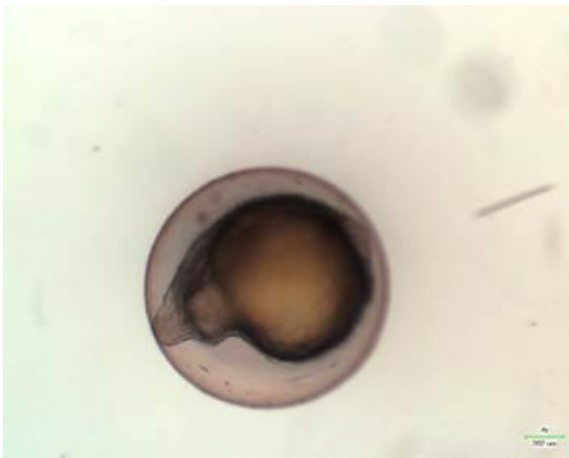


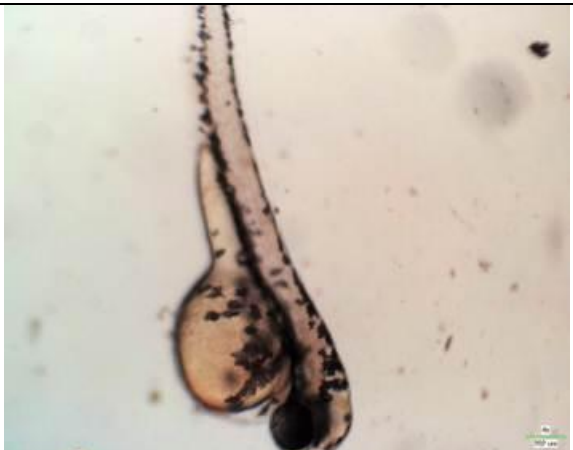
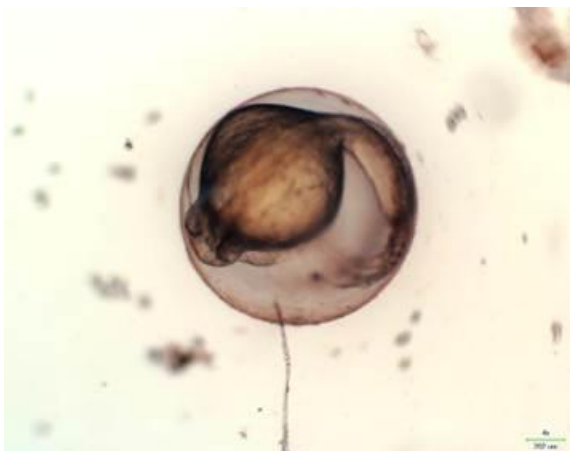
#### 4.8 Microscopic images of zebra fish embryos at different concentrations of CT



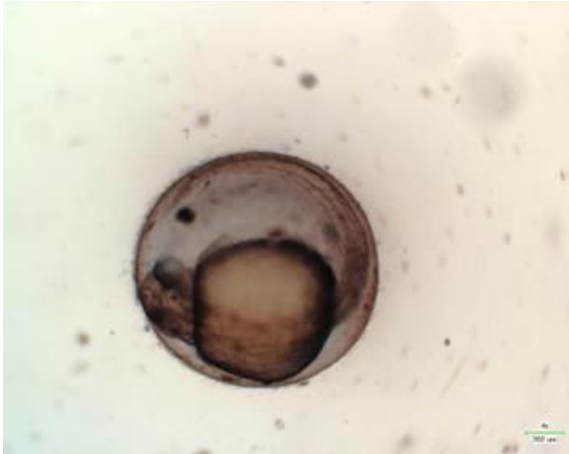
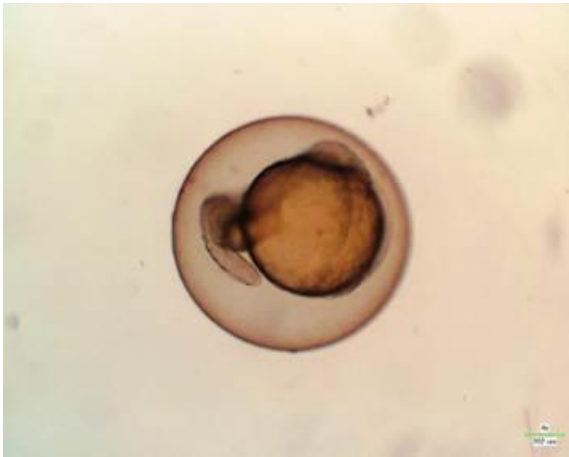
|                            | Microscopic Images                                                                   |
|----------------------------|--------------------------------------------------------------------------------------|
| <b>Control</b><br>24 hours |    |
| 48 hours                   |   |
| 72 hours                   |  |

|                                      |                                                                                      |
|--------------------------------------|--------------------------------------------------------------------------------------|
| <p><b>500 µg/ml</b><br/>24 hours</p> |    |
| <p>48 hours</p>                      |   |
| <p><b>250 µg/ml</b><br/>24 hours</p> |  |

|                              |                                                                                      |
|------------------------------|--------------------------------------------------------------------------------------|
| 48 hours                     |    |
| 72 hours                     |   |
| <b>100 μg/ml</b><br>24 hours |  |

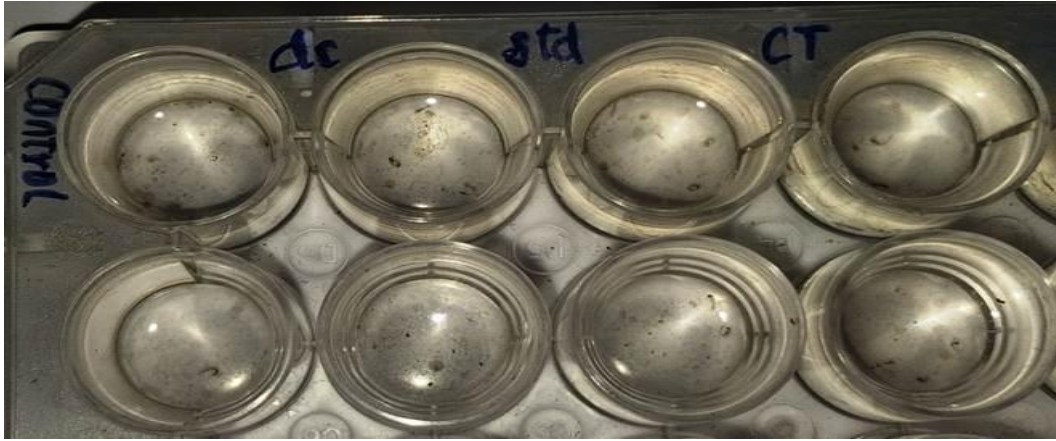
|                                 |                                                                                      |
|---------------------------------|--------------------------------------------------------------------------------------|
| 48 hours                        |    |
| 72 hours                        |   |
| 50 $\mu\text{g/ml}$<br>24 hours |  |



|                                                           |                                                                                      |
|-----------------------------------------------------------|--------------------------------------------------------------------------------------|
| <p>48 hours</p>                                           |    |
| <p>72 hours</p>                                           |   |
| <p><b>10 <math>\mu\text{g/ml}</math></b><br/>24 hours</p> |  |

|          |                                                                                     |
|----------|-------------------------------------------------------------------------------------|
| 48 hours |   |
| 72 hours |  |

#### 4.9 Effect of *Clitoria ternatea* extract on diabetes induced zebra fish larvae



|                                                | Microscopic Images                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Control                                        |  A microscopic image of a zebrafish embryo in the control group. The embryo is shown in a lateral view, with a clear, elongated body and a distinct head region. The yolk sac is visible as a large, clear, oval structure. The body is covered with numerous small, dark, pigmented spots, which are characteristic of zebrafish embryos. The background is a light, uniform color.               |
| Disease control<br>(Glucose- 16.7 $\mu$ M/ ml) |  A microscopic image of a zebrafish embryo treated with 16.7 $\mu$ M glucose. The embryo is shown in a lateral view, with a clear, elongated body and a distinct head region. The yolk sac is visible as a large, clear, oval structure. The body is covered with numerous small, dark, pigmented spots, which are characteristic of zebrafish embryos. The background is a light, uniform color. |
| Metformin (5 mg/ml)                            |  A microscopic image of a zebrafish embryo treated with 5 mg/ml metformin. The embryo is shown in a lateral view, with a clear, elongated body and a distinct head region. The yolk sac is visible as a large, clear, oval structure. The body is covered with numerous small, dark, pigmented spots, which are characteristic of zebrafish embryos. The background is a light, uniform color.   |

**Test sample (100 mg/kg)**



## 5.CONCLUSION AND FUTURE SCOPE

### 5.1 Conclusion

The present study evaluated the antidiabetic potential of *Clitoria ternatea* extract using a glucose-induced zebrafish embryo model. GC-MS analysis confirmed the presence of several phytochemical compounds that may contribute to the biological activity of the plant extract. The toxicity study helped determine suitable concentrations of the extract for further evaluation and indicated that the selected doses were appropriate for the experimental model.

The antidiabetic study showed that treatment with *Clitoria ternatea* extract produced a beneficial effect in glucose-induced zebrafish embryos when compared with the untreated diabetic group. These findings suggest that the phytochemicals present in the extract may play a role in reducing the effects of glucose-induced stress. The study also demonstrates that zebrafish embryos are a useful model for the preliminary screening of plant-based compounds with potential antidiabetic activity.

Overall, the results indicate that *Clitoria ternatea* possesses promising antidiabetic properties and may serve as a potential source of natural bioactive compounds. However, further investigations are necessary to confirm its efficacy and to understand the exact mechanism responsible for its therapeutic effects.

### 5.2 Future Scope

The findings of this study provide a basis for further research on the antidiabetic potential of *Clitoria ternatea*. Future studies can focus on isolating and characterizing the active phytochemical compounds responsible for the observed biological activity. Detailed molecular and biochemical studies may help to understand the mechanism through which the extract exerts its antidiabetic effect.

Further evaluation using adult zebrafish and other experimental animal models can provide additional evidence regarding its safety and efficacy. Long-term toxicity studies, dose optimization, and histopathological analyses may also be carried out to establish the therapeutic potential of the plant extract. In addition, formulation of the extract into suitable dosage forms and clinical investigations may be considered in the future to explore its possible application in the management of diabetes.

## 6.CO / PO ATTAINMENT TEMPLATE

This section is included to match the institutional style of the reference thesis. It may be edited as per the current department format and course code requirements.

| Course Outcome (CO) | Description                                                                                                                                       | Mapped PO       |
|---------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|
| CO1                 | Explain the basic concepts of diabetes, medicinal plants, and the use of zebrafish embryos in antidiabetic research.                              | PO1, PO2, PO4   |
| CO2                 | Perform GC-MS analysis of <i>Clitoria ternatea</i> extract and identify the major phytochemical compounds.                                        | PO3, PO4, PO5   |
| CO3                 | Evaluate the toxicity and antidiabetic activity of <i>Clitoria ternatea</i> using glucose-induced zebrafish embryos.                              | PO3, PO4, PO5   |
| CO4                 | Analyze the experimental results and interpret the antidiabetic potential of <i>Clitoria ternatea</i> with scientific and ethical considerations. | PO2, PO6, PO8   |
| CO5                 | Prepare a structured technical report and present the experimental findings effectively.                                                          | PO9, PO10, PO12 |

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